

S1 Checklist. TRIPOD+AI checklist for “Feature-set scaling for kidney failure prediction: development and internal validation of ten survival models against the Kidney Failure Risk Equation in MIMIC-IV.” Items follow Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement. BMJ 2024;385:e078378. “Portal” refers to the PLOS submission-form fields (funding, competing interests, data and code availability).

Item	Topic	Where reported / response
1	Title	Title: development and internal validation, kidney failure, MIMIC-IV.
2	Abstract	Abstract.
3a	Background	Introduction, paragraphs 1–2.
3b	Background	Introduction, paragraph 3: ranking hospitalized patients with CKD by risk of a recorded kidney-failure diagnosis, as an input to nephrology referral; research prototypes, not proposed for deployment.
3c	Background	Introduction, paragraph 2 (race coefficient in eGFR equations; race-free CKD-EPI 2021 used); Methods, Subgroup performance.
4	Objectives	Introduction, paragraph 3 and contributions (development and internal validation; comparison with KFRE).
5a	Data	Methods: Data source; Cohort. Development and evaluation data come from the same source (MIMIC-IV v2.2).
5b	Data	Methods: Data source (admissions 2008–2019; dates shifted per patient); Table 1 (follow-up).
6a	Participants	Methods: Data source (single centre, Beth Israel Deaconess Medical Center, Boston, USA; hospital and ICU records).
6b	Participants	Methods: Cohort; Outcome definition; Four and eight features; Twenty features.
6c	Participants	Methods: Cohort (no treatment information used as a predictor or for exclusion).
7	Data preparation	Methods: Four and eight features (matching rules, complete-case rows); Twenty features (value/missingness pairs); Resulting cohorts and data partitioning. Data quality was not assessed separately by sociodemographic group.
8a	Outcome	Methods: Outcome definition (ICD codes, event time, difference from KFRE’s endpoint; horizons of 2 and 5 years).
8b	Outcome	Not applicable: outcome ascertained from diagnosis codes, without assessor interpretation.
8c	Outcome	Methods: Outcome definition (no assessor blinding applies).
9a	Predictors	Introduction (scenario list); Methods: Four and eight features (KFRE inputs); Twenty features (20 most frequently measured labs).
9b	Predictors	Methods: Four and eight features (measurement timing relative to the anchor creatinine; uACR at or before the anchor; chemistry panel within ± 24 h); Twenty features.
9c	Predictors	Not applicable: laboratory values and recorded demographics.
10	Sample size	Methods: Cohort size (no formal calculation; size fixed by data availability).
11	Missing data	Methods: Four and eight features (complete case); Twenty features (missingness indicators).
12a	Analytical methods	Methods: Resulting cohorts and data partitioning (64/16/20 patient-level split, five seeds); Evaluation, Repetitions and aggregation.
12b	Analytical methods	Methods: Twenty features (predictors on recorded scales); KFRE baseline (published transformations).
12c	Analytical methods	Methods: Benchmarked models; Model training and optimization (hyperparameter search, selection criteria, validation data).
12d	Analytical methods	Not applicable: single centre; no clustering by site.
12e	Analytical methods	Methods: Evaluation (C-index, time-dependent AUC, integrated and fixed-horizon Brier scores, bootstrap intervals, calibration, decision curves, subgroup performance, feature importance).
12f	Analytical methods	Not applicable: no model updating or recalibration (Model training and optimization).
12g	Analytical methods	Methods: Evaluation, Prediction unit and Overall accuracy (native survival probabilities versus training-fitted conversion).
13	Class imbalance	Methods: Model training and optimization (no class-imbalance correction).
14	Fairness	Methods: Subgroup performance (performance within age, sex, and race groups; no fairness criterion defined or optimized); Discussion.

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15	Model output	Methods: Evaluation (risk scores and survival probabilities; decision thresholds 0.05–0.50 examined in decision curves; no single classification threshold proposed).
16	Training vs evaluation	Methods: Resulting cohorts and data partitioning (holdout is a random, outcome-stratified split of the same cohort); Table 1.
17	Ethical approval	Methods: Ethics statement.
18a	Funding	Portal: not applicable.
18b	Conflicts of interest	Portal: none declared.
18c	Protocol	No study protocol was prepared.
18d	Registration	The study was not registered.
18e	Data sharing	Methods: Data source; Portal (MIMIC-IV v2.2 via PhysioNet credentialed access, doi:10.13026/6mm1-ek67).
18f	Code sharing	Portal: code available in the project repository.
19	Patient and public involvement	None.
20a	Participants	Results: Participants; Table 1.
20b	Participants	Table 1 (age, sex, eGFR, uACR, follow-up, events); complete-case cohorts have no missing predictor values.
20c	Participants	Table 1 and Methods: Resulting cohorts and data partitioning (holdout drawn from the same pool, stratified on the event label).
21	Model development	Table 1 (train, test, and holdout patients; holdout events).
22	Model specification	Methods: KFRE baseline (full equations); Benchmarked models (architectures and hyperparameter ranges); code in the project repository.
23a	Model performance	Results: Tables 2–5 (means across splits, bootstrap 95% intervals, paired differences, subgroup performance); S1 and S2 Tables.
23b	Model performance	Not applicable: single centre.
24	Model updating	Not applicable: no model updating.
25	Interpretation	Discussion.
26	Limitations	Discussion: Limitations.
27a	Usability	Methods: Four and eight features (rows with missing inputs are excluded); Twenty features (missing values represented by indicators); Limitations.
27b	Usability	Not applicable: research prototypes; no user interaction is specified (Introduction).
27c	Usability	Discussion: Future work (external validation, prospective landmark, recalibration).